Q2 - Answers

1(a) Random demonstrations + Beam Search

We randomly sampled 6 different in-context sets from `optional_in_context_demonstrations` with sizes $k \in \{3, 4, 5, 6, 7, 8\}$. For each k, we built an in-context prompt and evaluated GPT2-medium on the TriviaQA validation set using beam search decoding.

k	Accuracy
3	0.130
4	0.105
5	0.145
6	0.130
7	0.080
8	0.120

Table 1: 1(a) Beam Search results on TriviaQA validation set for random demonstration sets of size k.

1(b) Random demonstrations + Sampling (Temperature = 0.7)

We repeated the same process as in 1(a), but replaced beam search with sampling decoding using temperature $T = 0.7$.

k	Accuracy
3	0.070
4	0.070
5	0.060
6	0.060
7	0.075
8	0.070

Table: 1(b) Sampling results on TriviaQA validation set for random demonstration sets of size k (temperature = 0.7).

Explanation: Beam search is deterministic and tends to select high-probability continuations, so its performance is more stable. Sampling introduces randomness (temperature = 0.7), which increases diversity but often produces off-topic or less factual completions, reducing accuracy.

2(a) Demonstrations Retrieval (BERT dot-product) + Evaluation

We encoded all 500 questions in `optional_in_context_demonstrations` using BERT embeddings. For each validation question, we retrieved the top-8 most similar demonstrations by dot product similarity and built a question-specific prompt, then evaluated GPT2-medium using beam search and sampling.

Decoding method	Accuracy
Beam search	0.120
Sampling ($T = 0.7$)	0.055

Table 2: 2(a) Retrieval-based in-context learning results on TriviaQA validation set.

2(b) Why does retrieval improve performance?

Retrieval improves performance because the selected demonstrations are semantically closer to the test question, so the prompt provides more relevant context and stronger hints about the expected answer type.

- **Less prompt noise:** Random demonstrations can be from totally different topics (like : sports, medicine, history), so the prompt feels mixed and GPT-2 may continue in the wrong direction. With retrieval, the examples are usually about similar content, so the prompt stays more focused.
- **Better guidance on the answer type:** Similar questions often require the same kind of answer (person, place, year, organization). For example:
  - If the retrieved questions start with "Who. . . ?", the demo answers are usually names.
  - If they start with "Which country. . . ?", the demo answers are usually country names.

  This makes GPT-2 more likely to output the correct *type* of answer instead of something unrelated.
- **Clearer few-shot pattern:** GPT-2 is not instruction-tuned, so it mainly learns what to do by copying patterns from the prompt. When we retrieve similar questions, the demonstrations look more consistent (topic, style, and answer format), so it is easier for the model to follow the "Question → short answer" behavior.
- **Better match to the grading function:** `check_answer_truthfulness` essentially checks whether one of the gold answer strings appears in the generated text. Retrieval-based prompts tend to produce shorter, more direct outputs (often just the entity), so the exact gold string is more likely to appear clearly, instead of being buried inside a long continuation.

3(a) Zero-shot on TriviaQA

We evaluated GPT2-medium in a zero-shot setting by prompting only with: `Question: {q} \nAnswer:` and measuring correctness using `check_answer_truthfulness`.

Decoding method	Accuracy
Beam search	0.085
Sampling ($T = 0.7$)	0.085

Table 3: 3(a) Zero-shot results on TriviaQA validation set.

3(b) Why are zero-shot results so low?

Zero-shot performance is low because GPT-2 is not instruction-tuned to answer questions directly. Without demonstrations, it often continues the prompt in an uncontrolled way (adding extra questions, generating unrelated narrative text, or producing long completions) instead of outputting a short factual answer, which reduces the chance of matching the gold answers.